Chapter 8

The Symplectic Group

Road map

The symplectic group is the linear symmetry group underlying classical Hamiltonian mechanics. We shall first write Hamilton's equations as one matrix equation. A change of phase-space coordinates is canonical precisely when its Jacobian preserves the corresponding skew-symmetric matrix. We then prove the canonical form and uniqueness of nondegenerate skew-symmetric matrices, establish the reciprocal pairing of symplectic eigenvalues and the identity $\det A = 1$, reformulate canonicality by Poisson brackets, and derive all canonical transformations of the form $q' = f(q)$.

8.1 Hamiltonian mechanics in matrix form

8.1.1 Phase space

Consider a mechanical system with n degrees of freedom. Its generalized coordinates and conjugate momenta are

$$q = (q_1, \ldots, q_n)^T, \qquad p = (p_1, \ldots, p_n)^T.$$

A physical state is represented by the phase-space point

$$x = (q, p)^T = (q_1, \ldots, q_n, p_1, \ldots, p_n)^T \in \mathbb{R}^{2n}.$$

As time varies, the system traces a phase-space trajectory

$$t \longmapsto x(t) = (q(t), p(t))^T = (x_1(t), \ldots, x_{2n}(t))^T.$$

The Hamiltonian

$$H = H(q, p) = H(x)$$

is the energy function that determines the motion. For example, a particle of mass m moving in three-dimensional Euclidean space in an external potential V has

$$H(q, p) = \sum_{i=1}^{3} \frac{p_i^2}{2m} + V(q_1, q_2, q_3).$$

Hamilton's canonical equations are

$$\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}, \qquad i = 1, \ldots, n. \tag{8.1}$$

8.1.2 The standard symplectic matrix

Let I_n denote the $n \times n$ identity matrix and define

$$J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}. \tag{8.2}$$

This $2n \times 2n$ matrix satisfies

$$J^T = -J, \qquad J^2 = -I_{2n}, \qquad J^{-1} = -J, \qquad \det J = 1.$$

Writing

$$\nabla_x H = \begin{pmatrix} \nabla_q H \\ \nabla_p H \end{pmatrix},$$

we obtain

$$J\nabla_x H = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix} \begin{pmatrix} \nabla_q H \\ \nabla_p H \end{pmatrix} = \begin{pmatrix} \nabla_p H \\ -\nabla_q H \end{pmatrix}.$$

Component by component,

$$[J\nabla H]_i = \begin{cases} \dfrac{\partial H}{\partial p_i}, & 1 \le i \le n, \\ -\dfrac{\partial H}{\partial q_{i-n}}, & n+1 \le i \le 2n. \end{cases}$$

Consequently, the $2n$ scalar equations (8.1) are exactly the single vector equation

$$\boxed{\dot{x} = J\nabla_x H(x).} \tag{8.3}$$

8.2 Coordinate representations and canonical transformations

8.2.1 The same physical state in two coordinate systems

It is useful to separate a physical state from the coordinates used to describe it. Let M be the phase-space manifold and let

$$X : M \supset U \longrightarrow X(U) \subset \mathbb{R}^{2n}, \qquad Y : M \supset U \longrightarrow Y(U) \subset \mathbb{R}^{2n}$$

be two coordinate charts. If $m = m(t) \in M$, define

$$x = X(m), \qquad y = Y(m).$$

The coordinate change is

$$y = f(x) = Y \circ X^{-1}(x), \qquad x = f^{-1}(y) = X \circ Y^{-1}(y).$$

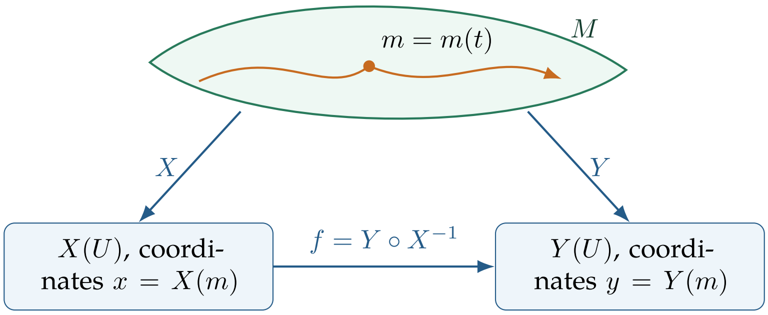


图 **8.1:** One phase-space point and two coordinate representations. The orange curve represents the same physical trajectory in M, independently of the chosen chart.

The Hamiltonian is intrinsically a scalar function $H : M \to \mathbb{R}$. Its two coordinate expressions are

$$H_X(x) = H(X^{-1}(x)), \qquad H_Y(y) = H(Y^{-1}(y)).$$

Because $y = f(x)$ implies $Y^{-1}(y) = X^{-1}(x)$, we have

$$H_X(x) = H_Y(f(x)) = H_Y(y). \tag{8.4}$$

Thus the numerical energy is unchanged; only its formula in coordinates may look different.

8.2.2 Deriving the Jacobian condition

Let $f : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ be a smooth invertible coordinate transformation and write

$$y = f(x) =: y(x).$$

Its Jacobian is

$$S(x) := Df(x) = \frac{\partial f}{\partial x}(x), \qquad S_{ij}(x) = \frac{\partial y_i}{\partial x_j}(x).$$

Since f is a local diffeomorphism, $S(x) \in GL(2n, \mathbb{R})$. At $y = f(x)$, the inverse Jacobian is

$$Df^{-1}(y) = S(x)^{-1}, \qquad \frac{\partial x_i}{\partial y_j} = [S(x)^{-1}]_{ij}.$$

Along a trajectory,

$$\dot{y}(t) = S(x(t))\dot{x}(t). \tag{8.5}$$

Moreover, differentiating (8.4) gives

$$\frac{\partial H_X}{\partial x_i} = \sum_{j=1}^{2n} \frac{\partial H_Y}{\partial y_j}\frac{\partial y_j}{\partial x_i}, \qquad \text{so} \qquad \nabla_x H_X = S(x)^T \nabla_y H_Y.$$

Substituting this and $\dot{x} = J\nabla_x H_X$ into (8.5) yields

$$\dot{y} = S(x)JS(x)^T \nabla_y H_Y. \tag{8.6}$$

Therefore the transformed equation has the same standard Hamiltonian form $\dot{y} = J\nabla_y H_Y$ exactly when

$$\boxed{S(x)JS(x)^T = J \qquad \text{for every } x.} \tag{8.7}$$

Definition 8.1: Canonical transformation

A smooth diffeomorphism f of phase space is called *canonical*, or a *symplectomorphism*, when its Jacobian satisfies (8.7) at every point. Such a transformation preserves the form of Hamilton's equations.

Why one also sees $S^T JS = J$

Our matrix $S_{ij} = \partial y_i/\partial x_j$ pushes coordinate velocities forward, and the direct calculation above produces $SJS^T = J$. If instead one describes the pullback of the bilinear symplectic form, the standard condition is $S^T JS = J$. For an invertible matrix and the standard J, the two conditions are equivalent: $S^T JS = J$ implies

$$S^{-1} = -JS^T J,$$

and hence $SJS^T = J$; the converse follows in the same way. The apparent transpose difference is therefore a matter of which direction of the coordinate change is being represented, not a difference in the geometry.

8.3 Definition and elementary group properties

More generally, let Γ be a fixed real, nondegenerate, skew-symmetric $2n \times 2n$ matrix:

$$\Gamma^T = -\Gamma, \qquad \det \Gamma \neq 0.$$

Definition 8.2: Symplectic group associated with Γ

$$\mathrm{Sp}(\Gamma) := \left\{ A \in GL(2n, \mathbb{R}) : A^T \Gamma A = \Gamma \right\}.$$

For the standard matrix J in (8.2), one writes

$$\mathrm{Sp}(2n, \mathbb{R}) = \mathrm{Sp}(J).$$

For the standard matrix $J^{-1} = -J$, the two equations

$$A^T JA = J \qquad \text{and} \qquad AJA^T = J$$

are equivalent. In particular $A \in \mathrm{Sp}(2n, \mathbb{R})$ if and only if $A^T \in \mathrm{Sp}(2n, \mathbb{R})$.

Proposition 8.1

$\mathrm{Sp}(\Gamma)$ is a group under matrix multiplication.

证明. Associativity is inherited from matrix multiplication, and $I_{2n}^T \Gamma I_{2n} = \Gamma$, so the identity belongs to the set. If $A_1, A_2 \in \mathrm{Sp}(\Gamma)$, then

$$(A_1 A_2)^T \Gamma (A_1 A_2) = A_2^T (A_1^T \Gamma A_1) A_2 = A_2^T \Gamma A_2 = \Gamma,$$

which proves closure. Finally, from $A^T \Gamma A = \Gamma$, multiply on the left by A^{-T} and on the right by

A^{-1}:

$$(A^{-1})^T \Gamma A^{-1} = \Gamma.$$

Thus $A^{-1} \in \mathrm{Sp}(\Gamma)$. ■

8.4 Canonical form of a nondegenerate skew matrix

Lemma 8.1: Orthogonal block form

Let Γ be a real nondegenerate skew-symmetric matrix. Then its dimension is even, say $2n$, and there is an orthogonal matrix $O \in O(2n)$ such that

$$O^T \Gamma O = \begin{pmatrix} 0 & \Lambda \\ -\Lambda & 0 \end{pmatrix}, \qquad \Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_n), \quad \lambda_j > 0. \tag{8.8}$$

Equivalently, after interleaving the basis vectors, the matrix is the block diagonal matrix

$$\mathrm{diag}\left(\begin{pmatrix} 0 & \lambda_1 \\ -\lambda_1 & 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 & \lambda_n \\ -\lambda_n & 0 \end{pmatrix} \right).$$

证明. First observe that

$$(\mathrm{i}\Gamma)^\dagger = -\mathrm{i}\Gamma^T = \mathrm{i}\Gamma.$$

Thus $\mathrm{i}\Gamma$ is Hermitian, so its eigenvalues are real. Consequently, the eigenvalues of Γ are purely imaginary. Since Γ has real entries, nonreal eigenvalues occur in conjugate pairs, and since $\det \Gamma \neq 0$, zero is not an eigenvalue. We may therefore list the eigenvalues as

$$\mathrm{i}\lambda_1, -\mathrm{i}\lambda_1, \dots, \mathrm{i}\lambda_n, -\mathrm{i}\lambda_n, \qquad \lambda_j > 0.$$

This already proves that the dimension is even.

Choose unit eigenvectors $u_j \in \mathbb{C}^{2n}$ such that

$$\Gamma u_j = \mathrm{i}\lambda_j u_j.$$

Complex conjugation gives

$$\Gamma \overline{u_j} = -\mathrm{i}\lambda_j \overline{u_j}.$$

The spectral theorem permits the vectors

$$u_1, \overline{u_1}, \dots, u_n, \overline{u_n}$$

to be chosen as a unitary basis of $\mathbb{C}^{2n}$. Define the real vectors

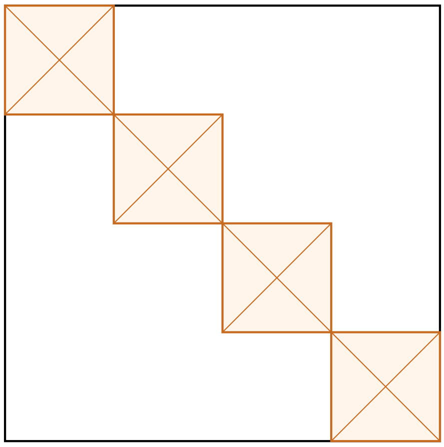
$$a_j = \frac{u_j + \overline{u_j}}{\sqrt{2}}, \qquad b_j = \frac{u_j - \overline{u_j}}{\sqrt{2}\,\mathrm{i}}.$$

They are real because $u_j + \overline{u_j} = 2\,\mathrm{Re}\,u_j$ and $(u_j - \overline{u_j})/\mathrm{i} = 2\,\mathrm{Im}\,u_j$. Unitarity gives

$$a_j^T a_k = \delta_{jk}, \qquad b_j^T b_k = \delta_{jk}, \qquad a_j^T b_k = 0.$$

Hence
$$O = (a_1, \ldots, a_n, b_1, \ldots, b_n)$$
is a real orthogonal matrix.

It remains to compute the action of Γ. From the eigenvector equations,
$$\Gamma a_j = \frac{\mathrm{i}\lambda_j u_j - \mathrm{i}\lambda_j \overline{u_j}}{\sqrt{2}} = -\lambda_j b_j,$$
$$\Gamma b_j = \frac{\mathrm{i}\lambda_j u_j + \mathrm{i}\lambda_j \overline{u_j}}{\sqrt{2}\,\mathrm{i}} = \lambda_j a_j.$$
Therefore
$$a_i^T \Gamma a_j = 0, \qquad a_i^T \Gamma b_j = \lambda_j \delta_{ij},$$
$$b_i^T \Gamma a_j = -\lambda_j \delta_{ij}, \qquad b_i^T \Gamma b_j = 0.$$
These four identities give precisely (8.8). ■



$$\mathrm{diag}\left(\begin{smallmatrix} 0 & \lambda_1 \\ -\lambda_1 & 0 \end{smallmatrix}, \ldots, \begin{smallmatrix} 0 & \lambda_n \\ -\lambda_n & 0 \end{smallmatrix}\right)$$

图 **8.2:** The 2×2 skew blocks in the interleaved real basis $(a_1, b_1, \ldots, a_n, b_n)$.

8.5 All nondegenerate symplectic forms give isomorphic groups

Lemma 8.2: Uniqueness up to congruence

Let Γ and Γ' be real nondegenerate skew-symmetric $2n \times 2n$ matrices. Then there is $Q \in GL(2n, \mathbb{R})$ such that
$$\Gamma = Q^T \Gamma' Q.$$
Moreover,
$$\mathrm{Sp}(\Gamma) \cong \mathrm{Sp}(\Gamma').$$

证明. By the orthogonal block-form lemma above, there are orthogonal matrices O, O' for which
$$D = O^T \Gamma O, \qquad D' = O'^T \Gamma' O'$$

are block diagonal with nonzero blocks

$$\begin{pmatrix} 0 & \lambda_j \\ -\lambda_j & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & \lambda'_j \\ -\lambda'_j & 0 \end{pmatrix}.$$

Choose a real invertible block diagonal matrix

$$R = \text{diag}(R_1, \ldots, R_n)$$

so that $R_j^T D'_j R_j = D_j$. For instance,

$$R_j = \begin{pmatrix} \sqrt{\lambda_j/\lambda'_j} & 0 \\ 0 & \sqrt{\lambda_j/\lambda'_j} \end{pmatrix}$$

works when the λ_j, λ'_j have been chosen positive. Hence

$$D = R^T D' R.$$

With

$$Q = O' R O^T$$

we obtain

$$Q^T \Gamma' Q = O R^T O'^T \Gamma' O' R O^T = O D O^T = \Gamma.$$

Now let $A \in \text{Sp}(\Gamma)$. Since $\Gamma = Q^T \Gamma' Q$,

$$Q^T \Gamma' Q = A^T Q^T \Gamma' Q A.$$

Multiplying by Q^{-T} on the left and Q^{-1} on the right gives

$$\Gamma' = (QAQ^{-1})^T \Gamma' (QAQ^{-1}).$$

Thus $QAQ^{-1} \in \text{Sp}(\Gamma')$. Conjugation

$$\Phi_Q : A \longmapsto QAQ^{-1}$$

is an automorphism of $GL(2n, \mathbb{R})$, and its inverse is conjugation by Q^{-1}. Its restriction is therefore the required group isomorphism. ■

In particular, every group defined by a real nondegenerate skew-symmetric form is isomorphic to the standard group

$$\text{Sp}(2n, \mathbb{R}) = \text{Sp}\left(\begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}\right).$$

8.6 Eigenvalues of a symplectic matrix

Theorem 8.1: Reciprocal and conjugate pairing

Let $A \in \mathrm{Sp}(\Gamma)$, where $\Gamma^T = -\Gamma$ and $\det \Gamma \neq 0$. If $\lambda_0 \in \mathbb{C}$ is an eigenvalue of A of algebraic multiplicity k, then

$$\overline{\lambda_0} \quad \text{and} \quad \lambda_0^{-1}$$

are also eigenvalues of algebraic multiplicity k. Consequently, $(\overline{\lambda_0})^{-1}$ has the same multiplicity as well.

证明. The symplectic identity gives

$$A^T \Gamma A = \Gamma \quad \Longrightarrow \quad A^{-T} = \Gamma A \Gamma^{-1}.$$

Let

$$\chi_A(\lambda) = \det(A - \lambda I_{2n})$$

be the characteristic polynomial in the convention used in the handwritten derivation. Similar matrices and transposed matrices have the same determinant, so

$$\begin{aligned}
\chi_A(\lambda) &= \det\bigl(\Gamma(A - \lambda I)\Gamma^{-1}\bigr) \\
&= \det(A^{-T} - \lambda I) \\
&= \det(A^{-1} - \lambda I).
\end{aligned}$$

Next,

$$\begin{aligned}
\det(A^{-1} - \lambda I) &= \det(A^{-1}) \det(I - \lambda A) \\
&= \det(A)^{-1} \lambda^{2n} \det(\lambda^{-1} I - A) \\
&= \det(A)^{-1} \lambda^{2n} \det(A - \lambda^{-1} I),
\end{aligned}$$

because the dimension $2n$ is even. Taking determinants in $A^T \Gamma A = \Gamma$ gives

$$(\det A)^2 \det \Gamma = \det \Gamma,$$

so at this stage $\det A = \pm 1$. Hence

$$\chi_A(\lambda) = \pm \lambda^{2n} \chi_A(\lambda^{-1}). \tag{8.9}$$

Suppose

$$\chi_A(\lambda) = (\lambda - \lambda_0)^k r(\lambda), \qquad r(\lambda_0) \neq 0.$$

Since A is invertible, $\lambda_0 \neq 0$. Substituting this factorization into (8.9) gives

$$\chi_A(\lambda) = \pm \lambda^{2n-k} (1 - \lambda\lambda_0)^k r(\lambda^{-1}).$$

Thus $\lambda = \lambda_0^{-1}$ is a zero of multiplicity at least k. Applying the same argument to λ_0^{-1} shows that the multiplicity cannot exceed k, so it is exactly k.

Finally, A is real, and therefore

$$\overline{\chi_A(\lambda)} = \chi_A(\overline{\lambda}).$$

Complex roots of a real polynomial occur with the same multiplicity as their complex conjugates. Hence $\overline{\lambda_0}$ is also a k-fold eigenvalue. ■

The four possible partners

Away from the real axis and the unit circle, one eigenvalue produces the quadruple

$$\lambda, \quad \overline{\lambda}, \quad \lambda^{-1}, \quad \overline{\lambda}^{-1}.$$

Some of these can coincide. For example, when $|\lambda| = 1$, $\lambda^{-1} = \overline{\lambda}$.

Proposition 8.2: A symplectic matrix has determinant 1

If $A \in \mathrm{Sp}(2n, \mathbb{R})$, then

$$\boxed{\det A = 1.}$$

Thus $\mathrm{Sp}(2n, \mathbb{R}) \subset SL(2n, \mathbb{R})$.

证明. For a fully rigorous argument, including the self-reciprocal eigenvalues $\lambda = \pm 1$, use the Pfaffian. For every $2n \times 2n$ skew-symmetric matrix B, the Pfaffian is a polynomial $\mathrm{pf}(B)$ satisfying

$$\mathrm{pf}(B)^2 = \det B, \qquad \mathrm{pf}(P^T BP) = \det(P)\,\mathrm{pf}(B).$$

The second identity follows directly by applying P to every argument of the alternating 2-form represented by B and comparing the coefficient of the top-degree volume form. Apply it to $A^T JA = J$:

$$\mathrm{pf}(J) = \mathrm{pf}(A^T JA) = \det(A)\,\mathrm{pf}(J).$$

Because J is nondegenerate, $\mathrm{pf}(J)^2 = \det J = 1$, so $\mathrm{pf}(J) \neq 0$. Cancellation gives $\det A = 1$.

The reciprocal-pair picture provides the same intuition: whenever $\lambda \neq \pm 1$, the factors λ and λ^{-1} multiply to 1. The Pfaffian argument also handles the fixed reciprocal points $\lambda = \pm 1$ and does not require A to be diagonalizable. ■

8.7 Examples of canonical transformations

Example 8.1: The identity

For $y = x$, the Jacobian is $S = I_{2n}$, and therefore

$$SJS^T = I_{2n} J I_{2n} = J.$$

The identity transformation is canonical.

Example 8.2: Exchange positions and momenta

Let

$$q' = p, \qquad p' = -q.$$

Then $y = Jx$, so $S = J$ is constant. Since $J^T = -J$ and $J^2 = -I$,

$$SJS^T = JJ(-J) = J.$$

Hence the transformation is canonical.

Example 8.3: Why $q' = f(q)$, $p' = g(q)$ cannot work

Suppose both new coordinates and new momenta depend only on the old coordinates. With

$$F = \frac{\partial f}{\partial q}, \qquad G = \frac{\partial g}{\partial q},$$

the Jacobian is

$$S = \begin{pmatrix} F & 0 \\ G & 0 \end{pmatrix}.$$

Its last n columns vanish, so S is singular; directly,

$$SJS^T = 0 \neq J.$$

Such a map discards all dependence on the old momenta and cannot be an invertible canonical transformation.

8.7.1 The first restrictions on a point transformation

Now take

$$q' = f(q), \qquad p' = g(q, p).$$

Put

$$F = \frac{\partial f}{\partial q}, \qquad G = \frac{\partial g}{\partial q}, \qquad P = \frac{\partial g}{\partial p}.$$

Then

$$S = \begin{pmatrix} F & 0 \\ G & P \end{pmatrix}, \qquad SJS^T = \begin{pmatrix} 0 & FP^T \\ -PF^T & GP^T - PG^T \end{pmatrix}.$$

The condition $SJS^T = J$ is therefore equivalent to

$$FP^T = I_n, \qquad GP^T - PG^T = 0. \tag{8.10}$$

The first equation gives

$$P = F^{-T}.$$

Since F depends only on q, integration with respect to p shows that p' must be affine in p:

$$p' = F(q)^{-T}p + h(q) \tag{8.11}$$

for some vector-valued function h. The second equation in (8.10) supplies the remaining integrability condition, which we derive more transparently by Poisson brackets below.

8.8 Poisson brackets

Definition 8.3: Poisson bracket

For smooth phase-space functions $A = A(q,p)$ and $B = B(q,p)$, define

$$\begin{aligned} \{A, B\} &= \sum_{i=1}^{n} \left(\frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - \frac{\partial B}{\partial q_i} \frac{\partial A}{\partial p_i} \right) && (8.12) \\ &= (\nabla A)^T J \nabla B. && (8.13) \end{aligned}$$

For $y = f(x)$, one has

$$\begin{aligned} \{y_i, y_j\} &= (\nabla_x y_i)^T J \nabla_x y_j \\ &= \sum_{k,\ell=1}^{2n} \frac{\partial y_i}{\partial x_k} J_{k\ell} \frac{\partial y_j}{\partial x_\ell} \\ &= (SJS^T)_{ij}. \end{aligned}$$

We have therefore proved the following convenient criterion.

Proposition 8.3: Canonical Poisson relations

A smooth invertible transformation $y = f(x)$ is canonical if and only if

$$\boxed{\{y_i, y_j\} = J_{ij} \qquad (1 \le i, j \le 2n).}$$

Writing $y = (q', p')$, this is equivalent to the four families

$$\begin{aligned} \{q'_i, q'_j\} &= 0, & \{q'_i, p'_j\} &= \delta_{ij}, \\ \{p'_i, q'_j\} &= -\delta_{ij}, & \{p'_i, p'_j\} &= 0. \end{aligned}$$

8.9 All canonical transformations with $q' = f(q)$

Assume $q' = f(q)$, with $F = \partial q'/\partial q$ invertible. Because q' has no p-dependence,

$$\{q'_i, q'_j\} = 0$$

automatically. The mixed relation gives

$$\delta_{ij} = \{q'_i, p'_j\} = \sum_{k=1}^{n} \frac{\partial q'_i}{\partial q_k} \frac{\partial p'_j}{\partial p_k}.$$

In matrix form,

$$F \left(\frac{\partial p'}{\partial p}\right)^T = I_n,$$

and hence

$$\frac{\partial p'}{\partial p} = F^{-T}.$$

This recovers

$$p' = A(q)p + h(q), \qquad A(q) = F(q)^{-T} = \left(\frac{\partial q}{\partial q'}\right)^T.$$

It remains to impose $\{p'_i, p'_j\} = 0$. Since

$$p'_i = \sum_m A_{im}(q) p_m + h_i(q),$$

we have

$$\frac{\partial p'_i}{\partial q_k} = \sum_m \frac{\partial A_{im}}{\partial q_k} p_m + \frac{\partial h_i}{\partial q_k}, \qquad \frac{\partial p'_i}{\partial p_k} = A_{ik}.$$

Therefore

$$\begin{aligned} \{p'_i, p'_j\} = \sum_{k,m} &\left(\frac{\partial A_{im}}{\partial q_k} A_{jk} - \frac{\partial A_{jm}}{\partial q_k} A_{ik}\right) p_m \\ &+ \sum_k \left(\frac{\partial h_i}{\partial q_k} A_{jk} - \frac{\partial h_j}{\partial q_k} A_{ik}\right). \end{aligned}$$

The coefficients of p_m cancel. To see the cancellation rather than merely assert it, use

$$A_{jk} = \frac{\partial q_k}{\partial q'_j}$$

and the chain rule:

$$\begin{aligned} \sum_k \frac{\partial A_{im}}{\partial q_k} A_{jk} &= \sum_k \frac{\partial}{\partial q_k}\left(\frac{\partial q_m}{\partial q'_i}\right) \frac{\partial q_k}{\partial q'_j} \\ &= \frac{\partial}{\partial q'_j}\left(\frac{\partial q_m}{\partial q'_i}\right) = \frac{\partial^2 q_m}{\partial q'_j \partial q'_i}. \end{aligned}$$

Interchanging i and j gives the same result because mixed partial derivatives commute. Thus only the h-terms remain:

$$0 = \{p'_i, p'_j\} = \sum_k \left(\frac{\partial h_i}{\partial q_k} \frac{\partial q_k}{\partial q'_j} - \frac{\partial h_j}{\partial q_k} \frac{\partial q_k}{\partial q'_i} \right)$$
$$= \frac{\partial h_i}{\partial q'_j} - \frac{\partial h_j}{\partial q'_i}.$$

This is precisely the local integrability condition for the one-form $\sum_i h_i \, \mathrm{d}q'_i$. On a simply connected coordinate neighbourhood, there is a scalar function S such that

$$h_i = \frac{\partial S}{\partial q'_i}, \qquad h = \nabla_{q'} S.$$

If the same scalar is expressed as a function of q, the chain rule gives

$$h_i = \sum_k \frac{\partial q_k}{\partial q'_i} \frac{\partial S}{\partial q_k} = [A(q) \nabla_q S(q)]_i.$$

Combining the results gives the complete local answer.

Theorem 8.2: Canonical point transformations

Let f be a local diffeomorphism of configuration space. The most general local canonical transformation whose new coordinates depend only on the old coordinates is

$$\boxed{\begin{aligned} q' &= f(q), \\ p' &= \left(\frac{\partial q}{\partial q'} \right)^T \big[p + \nabla_q S(q) \big], \end{aligned}}$$

where S is an arbitrary smooth scalar function. Equivalently,

$$p' = \left(\frac{\partial q'}{\partial q} \right)^{-T} \big[p + \nabla_q S(q) \big].$$

Local versus global statement

The equation $\partial h_i / \partial q'_j = \partial h_j / \partial q'_i$ guarantees a local potential S. A single global S exists under an additional topological hypothesis, for example on a simply connected domain. The handwritten computation is local, as are ordinary coordinate charts.

8.10 Reducing a general skew form to the standard form

Suppose the equations of motion are written with an arbitrary constant real nondegenerate skew-symmetric matrix Γ:

$$\dot{x} = \Gamma \nabla_x H(x). \tag{8.14}$$

By the uniqueness-up-to-congruence lemma, there is an invertible constant matrix Q such that

$$\Gamma = Q^T J Q.$$

Define new coordinates

$$y = Q^{-T} x, \qquad \text{so that} \qquad x = Q^T y.$$

If $H_Y(y) = H_X(Q^T y)$, then

$$\frac{\partial H_X}{\partial x_i} = \sum_j \frac{\partial H_Y}{\partial y_j} \frac{\partial y_j}{\partial x_i} = \sum_j (Q^{-T})_{ji} \frac{\partial H_Y}{\partial y_j},$$

and hence

$$\nabla_x H_X = Q^{-1} \nabla_y H_Y.$$

Using $\dot{y} = Q^{-T}\dot{x}$ in (8.14), we find

$$\begin{aligned}\dot{y} &= Q^{-T}\Gamma \nabla_x H_X \\ &= Q^{-T}(Q^T J Q) Q^{-1} \nabla_y H_Y \\ &= J \nabla_y H_Y.\end{aligned}$$

Thus every constant nondegenerate skew form can be brought to the standard Hamiltonian form by a linear change of variables.

8.11 Equations of motion as Poisson-bracket equations

For the coordinate function x_i, its gradient is the i-th standard basis vector e_i. Therefore

$$\{x_i, H\} = e_i^T J \nabla H = [J\nabla H]_i.$$

Hamilton's equation $\dot{x} = J\nabla H$ may consequently be written as

$$\boxed{\dot{x}_i = \{x_i, H\}, \qquad \text{or symbolically} \qquad \dot{x} = \{x, H\}.} \tag{8.15}$$

For $i = 1, \ldots, n$, this reads

$$\dot{q}_i = \{q_i, H\} = \frac{\partial H}{\partial p_i},$$

whereas for the momentum coordinate,

$$\dot{p}_i = \{p_i, H\} = -\frac{\partial H}{\partial q_i}.$$

Thus the bracket notation contains both families of Hamilton equations without losing any component information.

8.12 Geometric discussion and symmetry groups

Let $y = f(x)$ be canonical. If $x(t)$ solves

$$\dot{x} = J\nabla_x H_X(x),$$

then $y(t) = f(x(t))$ solves

$$\dot{y} = J\nabla_y H_Y(y).$$

The physical trajectory has not changed: it has merely been represented in different coordinates. One may therefore choose whichever representation makes the Hamiltonian or the equations easiest to analyze.

The invariant structure can be described in three equivalent languages:

1. the form of Hamilton's equations;
2. all Poisson brackets;
3. the antisymmetric bilinear symplectic form

$$\Omega(u, v) = u^T Jv.$$

With the tangent-vector convention, preservation of Ω means

$$\Omega(Su, Sv) = u^T S^T JSv = u^T Jv,$$

or $S^T JS = J$.

Definition 8.4: The full symplectomorphism group

The group of smooth canonical transformations of $\mathbb{R}^{2n}$ is

$$\mathrm{Symp}(\mathbb{R}^{2n}, J) = \left\{ f : \mathbb{R}^{2n} \to \mathbb{R}^{2n} : \begin{array}{l} f \text{ is a smooth diffeomorphism,} \\ Df(x)^T JDf(x) = J \text{ for every } x \end{array} \right\}.$$

This is an extremely large, infinite-dimensional Lie group. Its linear canonical subgroup is

$$\mathrm{Sp}(2n, \mathbb{R}) = \{S \in GL(2n, \mathbb{R}) : S^T JS = J\} \subset \mathrm{Symp}(\mathbb{R}^{2n}, J),$$

where the associated transformation is simply $f(x) = Sx$ and $Df(x) = S$ is constant.

Representation symmetry is not automatically a conservation law

A canonical coordinate transformation preserves the mathematical structure used to represent the dynamics. This alone does not say that a particular Hamiltonian is unchanged, so it does not by itself give a conserved quantity. A conservation law requires an actual symmetry of the Hamiltonian (together with the appropriate continuous one-parameter structure). Canonical symmetry in classical mechanics plays a role analogous to unitary changes of representation in quantum mechanics.

What to retain from this chapter

1. Hamilton's component equations combine into $\dot{x} = J\nabla H$, with $J = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$.
2. A phase-space diffeomorphism is canonical precisely when its pointwise Jacobian is symplectic; equivalently, it preserves all canonical Poisson brackets.
3. The linear symplectic group is $\mathrm{Sp}(2n, \mathbb{R}) = \{A : A^T J A = J\}$, and every one of its matrices has determinant 1.
4. Every real nondegenerate skew-symmetric matrix has even size and can be reduced to 2×2 skew blocks by an orthogonal basis. All resulting symplectic groups are isomorphic.
5. The spectrum of a real symplectic matrix is closed under complex conjugation and reciprocation, including algebraic multiplicity.
6. For $q' = f(q)$, every local canonical lift to phase space has
$$p' = \left(\frac{\partial q}{\partial q'}\right)^T [p + \nabla_q S(q)].$$
7. Canonical changes preserve the representation-independent Hamiltonian structure, but a change of representation is not by itself a physical conservation law.